

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

C02: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

Code of Conduct:

I AALH (1982050) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: AALH

Criteria	Weightage (%)	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Understanding of Scenario	20%	5	2	1	1
Identification of Key Issues	20%	1	1	1	1
Application of Concepts	25%	1	1	1	1
Decision Making	20%	1	1	1	1
Clarity of Response	15%	1	1	1	1

AALH 20/05/20

1) Smart campus (192.168.100.0/24)

Department	Subnet	Network Address	Host Range	Broadcast Address	Usable Hosts
School of Computing	/25	192.168.100.0	192.168.100.1 192.168.100.126	192.168.100.127	126
School of Electronics	/26	192.168.100.128	192.168.100.129 192.168.100.190	192.168.100.191	62
School of Mechanical Engineering	/27	192.168.100.192	192.168.100.193 192.168.100.253	192.168.100.255	30

Remaining unused IP range.

192.168.100.224 - 192.168.100.255

2) Software Company (10.10.0.0/16)

Department	Subnet	Network Address	Host Range	Broadcast Address	Usable Hosts
Software Development	/24	10.10.0.0	10.10.0.1 - 10.10.0.254	10.10.0.255	254
Quality Assurance	/25	10.10.1.0	10.10.1.1 - 10.10.1.126	10.10.1.127	126
Human Resources	/26	10.10.1.128	10.10.1.129 - 10.10.1.190	10.10.1.191	62
Executive Management	/27	10.10.1.192	10.10.1.193 - 10.10.1.255	10.10.1.255	30

Unused IP range:

10.10.1.224 - 10.10.255.255

3) IPv6

Given Address:

2001:0db8:abcd:0000:0000:0000:0000:0000/64

1) Compressed form:

2001:db8:abcd::/64

2) Expanded form:

2001:0db8:abcd:0000:0000:0000:0000:0000

3) Network prefix (/64)

2001:0db8:abcd:0000::/64

4) Hosts supported

2^{64}

5) Total possible Host Addresses:

18,446,744,073,709,551,616

4) Routing Table:

Routing Table

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

destination IP: 192.168.2.45

1) Destination subnet

192.168.2.0/24

2) Matching Routing Entry

192.168.2.0/24

3) Next - ~~the~~ Hop Network (Interface)

Interface connected to 192.168.2.0/24

4) Default Route (if no match)

0.0.0.0/0

5) Layer-3 Router

Administration LAN (192.168.10.0/26)

• Network Address: 192.168.10.0

• Host Range: 192.168.10.1 - 192.168.10.62

• Broadcast Address: 192.168.10.63

• Gateway: 192.168.10.1

Three computer IPs.

• 192.168.10.2

• 192.168.10.3

• 192.168.10.4

Finance LAN (192.168.10.64/26)

• Network Address: 192.168.10.64

• Host Range: 192.168.10.65 - 192.168.10.126

• Broadcast Address: 192.168.10.127

• Gateway: 192.168.10.65

Three computer IPs

• 192.168.10.66

• 192.168.10.67

• 192.168.10.68